

Linux

Awk

Awk Function	Description
<code>gsub(<i>r,s,t</i>)</code>	Globally substitutes <i>s</i> for each match of the regular expression <i>r</i> in the string <i>t</i> . Returns the number of substitutions. If <i>t</i> is not supplied, defaults to <code>\$0</code> .
<code>index(<i>s,t</i>)</code>	Returns position of substring <i>t</i> in string <i>s</i> or zero if not present.
<code>length(<i>s</i>)</code>	Returns length of string <i>s</i> or length of <code>\$0</code> if no string is supplied.
<code>match(<i>s,r</i>)</code>	Returns either the position in <i>s</i> where the regular expression <i>r</i> begins, or 0 if no occurrences are found. Sets the values of RSTART and RLENGTH .
<code>split(<i>s,a,sep</i>)</code>	Parses string <i>s</i> into elements of array <i>a</i> using field separator <i>sep</i> ; returns number of elements. If <i>sep</i> is not supplied, FS is used. Array splitting works the same way as field splitting.
<code>sprintf("<i>fmt</i>",<i>expr</i>)</code>	Uses printf format specification for expr .
<code>sub(<i>r,s,t</i>)</code>	Substitutes <i>s</i> for first match of the regular expression <i>r</i> in the string <i>t</i> . Returns 1 if successful; 0 otherwise. If <i>t</i> is not supplied, defaults to <code>\$0</code> .
<code>substr(<i>s,p,n</i>)</code>	Returns substring of string <i>s</i> at beginning position <i>p</i> up to a maximum length of <i>n</i> . If <i>n</i> is not supplied, the rest of the string from <i>p</i> is used.
<code>tolower(<i>s</i>)</code>	Translates all uppercase characters in string <i>s</i> to lowercase and returns the new string.
<code>toupper(<i>s</i>)</code>	Translates all lowercase characters in string <i>s</i> to uppercase and returns the new string.

Awk

Examples

```
$ cat table.txt  
brown bread mat hair 42  
blue cake mug shirt -7  
yellow banana window shoes 3.14
```

```
# print the second field of each input line
```

```
$ awk '{print $2}' table.txt  
bread  
cake  
banana
```

```
# print lines only if the last field is a negative number
```

```
# recall that the default action is to print the contents of $0
```

```
$ awk '$NF<0' table.txt  
blue cake mug shirt -7
```

```
# change 'b' to 'B' only for the first field
```

```
$ awk '{gsub(/b/, "B", $1)} 1' table.txt  
Brown bread mat hair 42  
Blue cake mug shirt -7  
yellow banana window shoes 3.14
```

https://learnbyexample.github.io/learn_gnuawk/